

SVETLOLOBOVO, Russian Federation

WMO: 296640

Lat: 55.0891N

Lon: 90.8485E

Elev: 325

StdP: 97.48

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.7	(c) -35.0	(d) -40.6	(e) 0.1	(f) -37.6	(g) -37.8	(h) 0.1	(i) -34.9	(j) 11.4	(k) -6.6	(l) 10.8	(m) -6.9	(n) 0.7	(o) 230	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.1	(c) 28.6	(d) 19.0	(e) 26.9	(f) 18.2	(g) 25.2	(h) 17.4	(i) 20.4	(j) 25.9	(k) 19.4	(l) 24.8	(m) 18.5	(n) 23.7	(o) 2.5	(p) 50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.9	22.6	17.5	13.0	21.7	16.6	12.3	20.8	60.3	26.2	56.8	25.1	54.0	24.0	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.2	(b) 8.6	(c) 7.4	(e) -39.1	(f) 31.7	(g) 3.4	(h) 1.8	(i) -41.6	(j) 33.0	(k) -43.6	(l) 34.0	(m) -45.5	(n) 35.0	(o) -48.0	(p) 36.3		
(4) 10.2	(5) 8.6	(6) 7.4	(7) N/A	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-18.8	-16.4	-6.5	3.3	8.8	16.6	18.0	15.9	8.8	1.5	-9.3	-13.9		
	DBStd	14.27	9.76	8.84	7.35	4.57	4.92	3.91	2.59	3.06	4.09	5.10	7.87	8.90		
	HDD10.0	4092	893	740	511	206	84	4	0	1	69	263	580	740		
	HDD18.3	6474	1151	973	769	450	297	74	38	86	286	521	830	998		
	CDD10.0	725	0	0	0	6	47	203	249	184	34	1	0	0		
	CDD18.3	66	0	0	0	0	2	24	28	11	1	0	0	0		
	CDH23.3	817	0	0	0	2	34	353	284	130	13	0	0	0		
	CDH26.7	161	0	0	0	0	4	81	53	21	1	0	0	0		
	WSAvg	2.7	3.0	2.5	2.8	3.2	3.7	2.1	1.9	1.8	2.3	3.1	3.2	3.1		
	PrecAvg	361	8	10	9	17	28	61	65	67	39	23	19	15		
(15) Precipitation	PrecMax	452	17	30	21	46	62	119	135	196	89	50	37	35		
	PrecMin	255	2	1	2	2	8	30	18	19	13	8	6	2		
	PrecStd	55	4	7	5	11	13	20	31	36	17	10	8	9		
	PrecStd	55	4	7	5	11	13	20	31	36	17	10	8	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	2.1	11.0	20.8	26.7	31.0	30.3	29.0	25.3	16.5	8.9	2.2		
		MCWB	-1.4	0.7	5.4	11.8	14.5	18.9	19.6	21.0	16.6	10.7	6.3	-0.2		
	2%	DB	-1.3	0.1	7.7	16.9	23.6	29.0	28.3	26.6	21.2	13.4	5.1	0.2		
		MCWB	-3.1	-1.7	3.4	9.0	13.1	18.6	19.1	19.2	15.5	8.7	2.5	-1.7		
	5%	DB	-3.6	-1.5	5.4	14.2	21.0	27.1	26.5	24.5	18.4	11.1	2.7	-1.1		
		MCWB	-5.4	-3.2	2.1	7.9	11.8	17.9	18.8	18.0	13.8	7.4	0.5	-2.6		
	10%	DB	-5.3	-3.4	3.4	11.9	18.2	25.3	24.8	22.4	15.8	9.1	0.6	-2.8		
		MCWB	-6.9	-4.8	0.8	6.5	11.0	16.8	18.2	17.0	12.3	5.7	-1.1	-4.2		
	0.4%	WB	-1.0	0.7	5.9	11.8	15.8	21.0	21.4	21.7	17.3	11.0	6.2	0.0		
		MCDB	0.6	1.9	10.3	20.1	23.1	27.8	26.6	27.5	22.9	15.1	9.0	1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-3.2	-1.4	3.8	9.7	14.3	19.8	20.3	20.0	15.8	9.3	2.5	-1.3		
		MCDB	-1.2	-0.2	7.1	15.3	21.2	26.1	25.5	25.2	20.3	12.9	4.7	0.1		
	5%	WB	-5.3	-3.0	2.2	8.3	12.8	18.8	19.6	18.7	14.3	7.7	0.7	-2.6		
		MCDB	-3.8	-1.5	5.0	13.7	19.0	25.2	24.9	22.8	18.0	10.8	2.6	-1.1		
	10%	WB	-6.9	-5.0	0.9	6.7	11.5	17.9	18.7	17.8	12.7	6.1	-0.8	-4.1		
		MCDB	-5.4	-3.4	3.3	11.6	17.9	24.2	24.0	21.6	15.4	8.8	0.6	-2.6		
	5% DB	MDBR	8.1	9.3	10.6	11.1	11.4	12.7	11.1	11.0	10.4	8.2	7.6	7.8		
		MCDBR	7.2	8.1	10.5	14.8	17.3	16.4	14.7	14.5	15.4	12.3	6.8	5.5		
		MCWBR	6.3	7.0	7.2	8.5	9.0	7.3	6.9	7.5	9.4	8.0	5.1	4.8		
		MCWBR	7.0	7.1	9.9	14.0	14.5	13.9	12.5	12.4	14.5	11.9	6.8	5.5		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.1	6.1	7.0	8.6	8.2	6.6	6.5	6.9	9.1	8.0	5.4	5.0		
		MCWBR	6.1	6.1	7.0	8.6	8.2	6.6	6.5	6.9	9.1	8.0	5.4	5.0		
	taub	taub	0.277	0.312	0.360	0.401	0.410	0.405	0.427	0.410	0.370	0.347	0.312	0.279		
		taud	2.035	2.008	1.946	2.092	2.185	2.301	2.246	2.262	2.314	2.236	2.056	1.974		
(40) Clear-Sky Solar Irradiance	Ebn at Noon	Ebn at Noon	617	733	782	813	832	841	811	800	784	697	557	501		
		Edh at Noon	68	102	139	139	135	122	126	117	97	82	67	58		
	RadAvg	RadAvg	0.88	1.88	3.33	4.46	5.05	5.96	5.28	4.35	2.93	1.56	0.77	0.55		
(44) All-Sky Solar Radiation	RadStd	RadStd	0.09	0.15	0.31	0.25	0.57	0.51	0.30	0.39	0.32	0.15	0.10	0.07		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page